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Climate change includes both the global warming driven by human emissions of greenhouse gases, and the resulting large-scale shifts in weather patterns. Though there have been previous periods of climatic change, since the mid-20th century the rate of human impact on Earths climate system and the global scale of that impact have been unprecedented.

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That human activity has caused climate change is not disputed by any scientific body of national or international standing. The largest driver has been the emission of greenhouse gases, of which more than 90% are carbon dioxide and methane. Fossil fuel burning for energy consumption is the main source of these emissions, with additional contributions from agriculture, deforestation, and industrial processes.

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Temperature rise is accelerated or tempered by climate feedbacks, such as loss of sunlight-reflecting snow and ice cover, increased water vapour (a greenhouse gas itself), and changes to land and ocean carbon sinks. Impacts of climate change include warming temperatures, changes in precipitation, and increases in the frequency or intensity of some extreme weather events.